Yuheng Tao

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Research Interests

Wearable sensing and physical human-computer interaction, focusing on low-burden physiological and behavioral sensing to understand user states and provide intuitive embodied feedback and health support, with further interests in personalized and adaptive interaction informed by measurable human responses.

Education

National University of Singapore *Aug 2025 - Jan 2027*

Master of Science in Engineering Design and Innovation

GPA: 3.94/5

Xi'an Jiaotong-Liverpool University *Sep 2020 - Jul 2024*

BEng (Hons) in Industrial Design, First Class

Academic Honours: Academic Excellence Scholarship (Top 5%, 2023/24); Academic Achievement Scholarship (2021/22)

Publications

Yuheng Tao, Xialu Chen, and Huayi Liu. 2026. Breatho: Real-Time Respiratory Guidance System for Amateur Runners. In *Companion Publication of the 2026 ACM Designing Interactive Systems Conference* (DIS Companion '26), June 13-17, 2026, Singapore, Singapore. ACM, New York, NY, USA.

Yuheng Tao, Tingxiao Wu, and Shaoyu Cai. 2026. Demonstration of Breatho: Cadence-Synchronized Breathing Guidance for Amateur Runners. In *SIGGRAPH Asia 2026 Emerging Technologies* (SA Emerging Technologies '26), December 1-4, 2026, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 2 pages. (Accepted).

Huzhaorui Kang, Zhuolun Li, Tingxiao Wu, **Yuheng Tao**, Quanhao Gan, Tao Lin, Kongpyung (Justin) Moon, and Shaoyu Cai. 2026. PneuStep: A Soft Pneumatic-based Foot Haptic Device for Walking Experience on Different Terrains in VR. In *SIGGRAPH Asia 2026 Emerging Technologies* (SA Emerging Technologies '26), December 1-4, 2026, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 3 pages. (Accepted).

Research Experience

Capacitive Touch Super-Resolution | *Research Project & Manuscript in Preparation* *Mar 2026 - Present*

- **Designed and fabricated a low-channel capacitive sensing prototype** for fine-grained touch localization, investigating how spatial positions produce distinguishable response patterns across a limited number of sensing channels.
- **Developed an end-to-end touch localization pipeline**, independently completing electrode geometry design, sensing hardware integration, signal acquisition, and K-NN-based position classification; implemented a greedy algorithm to automatically generate electrode layouts that improve spatial distinguishability across channels.
- **Achieved 99.1% position classification accuracy** across 80 test locations spaced 5mm apart within a 50mm-diameter continuous sensing area, demonstrating millimeter-scale position discrimination with a low channel count.
- **Evaluating spatial resolution limits and generalizability** through finer sampling intervals and controlled comparisons across electrode layouts, baseline methods, and surface geometries.

Supervisor: Dr. Shaoyu Cai | Immersive Reality Lab, National University of Singapore

Breatho: Cadence-Synchronized Pneumatic Breathing Guidance | *Project Lead & First Author* *Aug 2025 - Present*

- **Initiated a five-person wearable-haptics project** that coupled breathing guidance with running cadence, developing a wearable system that delivers periodic abdominal pneumatic feedback during running without continuous visual or auditory attention.
- **Contributed substantially to three hardware iterations**, progressing from an IMU-based MVP to a dual-pump, dual-valve, dual-ESP32 architecture; worked on firmware adjustment, pressure-sensor calibration, and electro-pneumatic integration, and optimised signal filtering, pressure control, and actuation timing for reliable real-time feedback during movement.
- **Produced two first-author publications** at ACM DIS 2026 Companion and SIGGRAPH Asia 2026 Emerging Technologies; led the writing, submission, camera-ready preparation, and presentation materials for the latter.

Supervisor: Dr. Shaoyu Cai, National University of Singapore

Multimodal Haptics and Human Perception | *Research Assistant* *Aug 2026 - Present*

- **Conducted controlled user experiments on multimodal haptic perception**, investigating interactions among physical stimuli and quantifying perceptual responses across different stimulus conditions.
- **Stabilized and debugged the experimental system** through sensor calibration, hardware troubleshooting, and optimization of multimodal haptic actuation modules; contributed to protocol design and pilot iterations, and independently managed participant recruitment and 128 controlled trials per participant.
- **Processed and analyzed experimental data using repeated-measures ANOVA (RM-ANOVA)** to examine the effects of stimulus conditions on perceptual measures; **currently responsible for manuscript revision**, including experiments and additional data analysis.
- *Supervisor: Dr. Shaoyu Cai | Immersive Reality Lab, National University of Singapore*

Selected Projects

CartX: Smart Airport Mobility and Luggage Trolley | *Human Factors & Structural Design* *Aug 2025 - Apr 2026*

- **Conducted field research, questionnaires, and user-needs analysis** with travelers carrying varying amounts of luggage, translating findings into a variable-capacity storage system, sliding compartment doors, and chassis connection structures designed in Creo.
- **Fabricated and tested laser-cut and 3D-printed prototypes** through seven follow-and-navigation trials along a 50 m indoor route, iterating the structure based on test results; received the Best Poster Award at the NUS EDIC Project Showcase 2026.

Hydroren: Hydrocyclone Robotic Pool Cleaner | *Engineering Simulation & System Optimization* *Sep 2023 - Apr 2024*

- **Proposed and designed a filter-free robotic pool-cleaning system** using hydrocyclone solid–liquid separation to address clogging and frequent maintenance in conventional filter-based cleaners; independently developed the system architecture, internal component layout, and full assembly model.
- **Performed fluid–particle simulations in ANSYS Fluent** to evaluate inlet geometries, operating pressures, and particle sizes, iterating from a single hydrocyclone to a 10-hydrocyclone array; achieved simulated separation efficiencies of 100% for ≥100 μm particles and 94.2% for 50 μm particles, with theoretical power consumption ≤21 W.

Beacon: Computer-Vision Adaptive Dining Light | *Intelligent Interaction* *Jun 2023 - Apr 2024*

- **Independently designed and built a Raspberry Pi-based adaptive dining light** using computer vision to detect tableware positions and control four lighting zones, enabling lighting to respond dynamically to changes in the dining environment.
- **Collected and annotated approximately 200 images, trained the recognition model, and integrated the hardware and software into a functional prototype**, achieving 98% initial recognition accuracy; exhibited the project at SaloneSatellite 2024 in Milan and the 53rd China International Furniture Fair (Guangzhou).

Teaching Experience

Teaching Assistant | *CDE5312 Beyond the Surface: Exploring Interactive Wearables* *Aug 2026 - Present*

- Supported hands-on teaching in wearable interaction, guiding students through experiments and prototyping while providing technical support and troubleshooting for sensor integration, circuit connections, embedded control, and structural implementation.

Research & Technical Skills

Research & Experimentation: Controlled user studies, interviews, questionnaires, user testing, experimental data analysis, RM-ANOVA, K-NN classification

Hardware & Prototyping: Arduino, sensor integration, circuit soldering, pneumatic systems, experimental apparatus debugging, 3D printing, laser cutting

Research Tools: Matlab, Rhino, Creo, Blender, KeyShot, Figma, Adobe Photoshop/Illustrator, Zotero

Languages: Mandarin Chinese (native), English (fluent; IELTS 7.0), Japanese (basic)